Towards Affordable Ultrasound Wearables with Polymer-Based CMUTs

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Background, Motivation and Objective

Polymer-based Capacitive Micromachined Ultrasonic Transducers (polyCMUTs) are a promising acoustic frontend technology for next-generation ultrasound systems. Their high receive sensitivity, design freedom, compatibility with rapid, low-cost fabrication and their ability to form conformal, small-scale arrays make them particularly attractive for wearables. This work presents recent progress with polyCMUTs, focusing on device performance as well as integration with ultra-low-power electronics to enable customizable, miniaturized, and affordable ultrasound wearables for varying applications.

Statement of Contribution/Methods

State-of-the-art polyCMUT devices were designed and fabricated in-house using refined and optimized surface micromachining processes to produce structural polymer layers. Transducer arrays with 2, 8, and 16 channels were developed for a range of biomedical and industrial monitoring applications. For system integration, the Wearable Ultra-Low-Power Ultrasound (WULPUS) platform was adapted to support different configurations. The resulting systems provide ±30 V DC bias, configurable excitation pulses up to 30 V, adjustable channel counts, and lightweight, battery-compatible operation.

Results/Discussion

Figure (a) presents several recent prototypes: (left) an 8-channel pulse-echo system for industrial weld inspection and small-ROI imaging; (center) a low-cost (<USD 25) two-channel system with separate TX/RX paths enabling continuous 50 Hz monitoring for more than 40 hours on battery power; and (right) a highly miniaturized pulsed-wave Doppler system powered by a CR2032 coin cell with up to 70 hours of continuous operation. For the latter system, we further explored the geometric design flexibility of polyCMUTs to implement a ring-based, hexagonal TX-RX layout that produces quasi-isotropic and spatially co-aligned acoustic beam profiles for both channels (Fig. (b)). This architecture eliminates the need for high-voltage switching to reduce crosstalk while enabling quasi omnidirectional operation. Overall, the recent results demonstrate the potential of polyCMUTs to enable customizable, compact, low-cost, and energy-efficient ultrasound wearables for continuous monitoring applications.